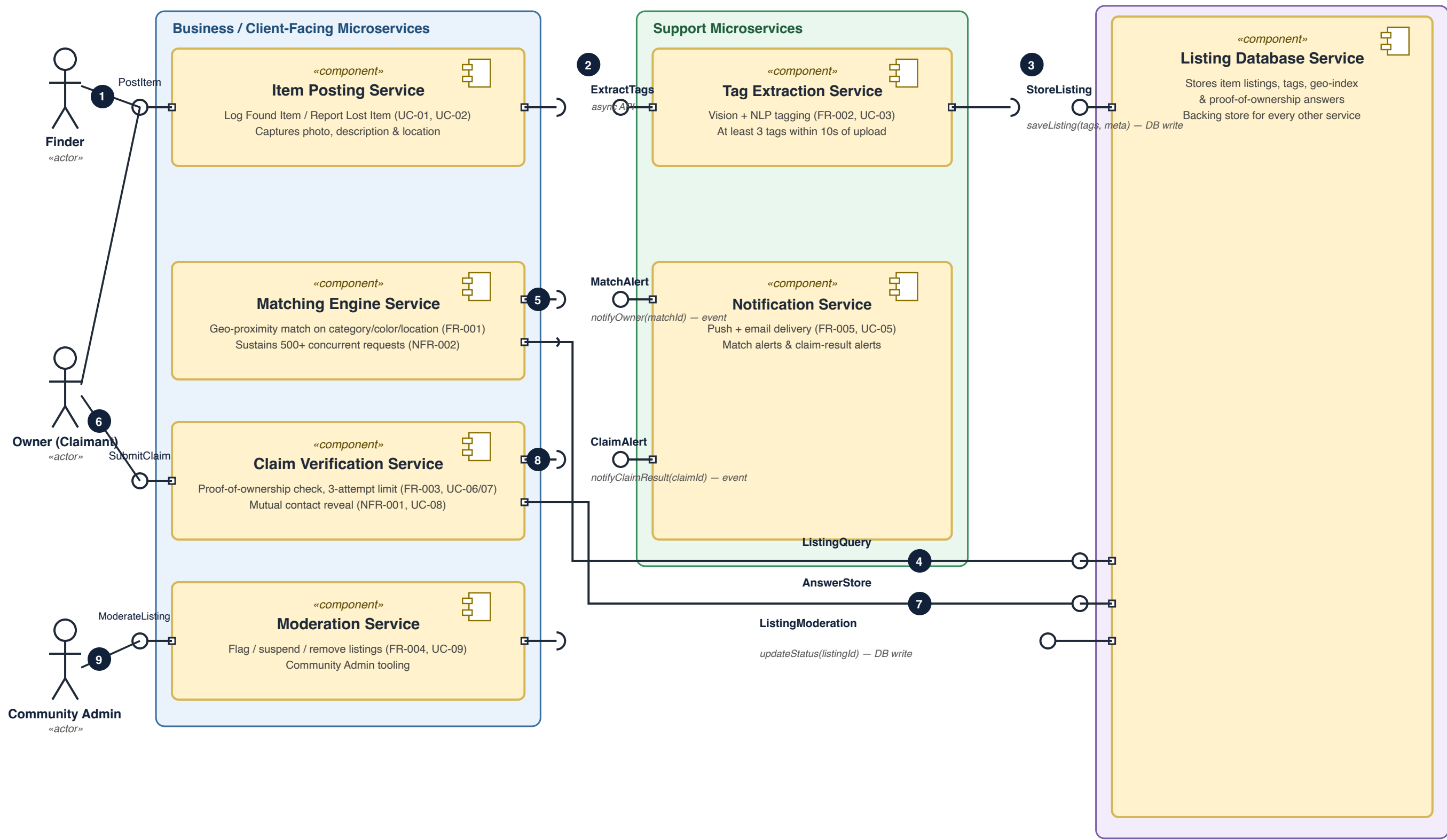


Community Lost & Found Matching Network — UML Component Diagram

Microservices Architecture — Lab 3 (PES1UG24CS929, Problem Statement #59)



Microservices Principle

Each box is an independently deployable service that owns its own scaling policy.

The Matching Engine Service is scaled out to absorb posting-time traffic spikes (NFR-002) without over-provisioning Moderation or Notification, which run at far lower, steadier load. Services communicate only through the labelled provided/required interfaces above; none share another service's database.

Claim Verification Service is isolated behind its own interface so that a compromise elsewhere cannot reach stored answers or contact details.

Data Flow — Main Scenarios

- 1 Finder/Owner posts a found or lost item (UC-01/UC-02) via `PostItem`.
- 2-3 Item Posting Service requests tags, which are extracted and saved (FR-002, UC-03).
- 4 Matching Engine queries the Listing Database for geo/category candidates (FR-001).
- 5 On a match above threshold, the Owner is alerted (FR-005, UC-05).
- 6 Owner submits a claim and answers the Finder's proof-of-ownership question (UC-06).
- 7 Claim Verification checks the answer against the Listing Database (FR-003, UC-07).
- 8 Both parties are notified and contact details mutually revealed (NFR-001, UC-08).
- 9 Community Admin may flag/suspend/remove any listing at any time (FR-004, UC-09).

Notation

- Provided interface (ball)
- ⌋ Required interface (socket)
- Port
- ➔ Direction of request
- Step in main data flow
- «component» (microservice)